Which LMS Features Actually Predict Student Attainment?

A predictive analysis of learning management system engagement data

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1. Introduction

Learning Management Systems (LMS) such as Moodle, Canvas, and Blackboard now sit at the centre of higher-education delivery, generating an enormous trail of behavioural data (Siemens and Baker, 2012). The promise of learning analytics is that this data can distinguish between features merely clicked and those that carry meaningful information about learning. The reality is messier. Studies reporting strong links between engagement and grades (You, 2016) sit alongside others arguing that raw activity counts are too crude to be trusted (Gaš ević , Dawson and Siemens, 2015). For an instructor deciding which LMS features to keep and which to drop, that disagreement directly affects what students experience next term.

This proposal responds to the second topic area in the module brief: A course instructor has introduced several new LMS features and now wants to know which have independent predictive value for student grades. The instructor has provided anonymised engagement records for the current cohort in CSV form, final grades for that cohort, and the previous cohort's grades from a year before the new features were introduced. The brief sets out a working hypothesis (specific features are associated with higher attainment, alone or in combination) against a sharp null hypothesis (any apparent effect is attributable solely to total access volume; students who click more get higher grades, regardless of what they click on).

Distinguishing those two possibilities is harder than it looks. LMS engagement variables are notoriously intercorrelated: a student who watches lecture videos also tends to complete quizzes, post on the forum, and download slides (Tempelaar, Rienties, and Giesbers, 2015). A naive regression on raw counts will conclude that everything matters when a single underlying factor, call it conscientiousness or selfregulation, is doing most of the work. The methodological challenge is to demonstrate that specific features predict grades beyond a generic effort signal.

The research question is therefore: “Which engineered LMS engagement features make an independent contribution to predicting student grades, after controlling for total platform access?” The design supports statements about predictive contribution after controlling for plausible confounders, not causal claims about feature effectiveness.

2. Background & Related Work

2.1 Engagement and attainment: what the evidence does, and does not show

Learning analytics, defined by Siemens and Baker (2012) as the use of educational data to better understand and support learning, has produced two largely separate strands of evidence. The first shows that students who engage more on the LMS tend to achieve higher average grades (Macfadyen and Dawson, 2010; AgudoPeregrina et al., 2014). The second asks whether this average correlation reflects a causal effect of any feature, or simply the fact that conscientious students do more of everything (Gaš ević , Dawson and Siemens, 2015).

Macfadyen and Dawson (2010) built one of the first large-scale logistic regression models using LMS variables to predict grade outcomes, finding that just four behavioural variables, namely total time online,

forum posts, email messages, and assessments completed, carried most of the predictive load. AgudoPeregrina et al. (2014) replicated this across three course contexts, showing that the predictive value of LMS variables is not stable across courses; what works in a fully online module looks different in a blended one. The implication is direct: a study designed to inform one instructor's decisions must work with that instructor's data on its own terms, rather than borrowing conclusions from larger, crossinstitutional studies.

2.2 Why raw click counts are not enough

The methodological critique by Gaš ević , Dawson and Siemens (2015) is that raw engagement counts conflate three distinct things: meaningful learning behaviour, low-effort clicking, and accidental signals (such as a video left running in another tab). Pardo, Han and Ellis (2017) demonstrated this by combining LMS trace data with self-reported learning strategies, finding that the same engagement behaviour predicted grades positively for some students and negatively for others. Ló pez-Pernas, Saqr and Viberg (2023) reframe the problem in multidimensional terms: a single engagement score collapses what should be several behavioural dimensions, and predictions based on the collapsed score systematically miss patterns that disaggregated data would surface.

This matters because it explains why simple correlation analysis is insufficient and why the ML approach proposed in Section 3 is appropriate. If the relationship between features and grades is non-linear and involves feature interactions, a linear model will under-represent the true predictive structure. Ensemble methods such as Random Forest and gradient boosting handle non-linearities and interactions natively (Hastie, Tibshirani and Friedman, 2009; Saqr and Ló pez-Pernas, 2021), which is the substantive reason for choosing them.

2.3 Methodological lessons from prior LMS-prediction studies

Three methodological choices recur across well-cited LMS-prediction studies and shape the design here. First, prior attainment matters. Owston, Lupshenyuk and Wideman (2011) found that the apparent benefit of lecture capture was concentrated among students who were already strong; without prior attainment controls, the effect appears larger and more universal than it is. The CSV data does not include prior grades directly, but the previous cohort's grade distribution serves as a population-level baseline.

Multicollinearity must be addressed, not ignored. Tempelaar, Rienties and Giesbers (2015), using a dataset of 909 students, showed that engagement variables typically share 40–70% of their variance and that ordinary regression on raw counts yields unstable coefficients. Their recommended remedies are Lasso regression (Tibshirani, 1996) and principal component reduction; both are drawn on directly from a small toolkit used in this project, with Variance Inflation Factor diagnostics (Kutner et al., 2005) used to trigger the switch.

No single algorithm is sufficient. Conijn et al. (2017), comparing logistic regression and random forests across seventeen Moodle courses, found that the methods agreed on the strongest predictors but diverged on weaker ones. Reporting both is methodologically sound: regression yields interpretable coefficients with confidence intervals, while ensemble methods yield more stable variable-importance rankings under multicollinearity (Strobl et al., 2008). This constitutes triangulation across two methods with complementary failure modes.

2.4 The gap this project addresses

Three observations follow. First, prior LMS-prediction work has relied on very large MOOC datasets or cross-institutional comparisons; how a single instructor with one dataset should decide what to keep, drop, or augment has been studied less. Second, where multicollinearity is acknowledged, it is rarely addressed with appropriate statistical methods; ordinary regression remains the default. Third, modern educational ML studies often report a single black-box accuracy score without explaining which features drove the prediction (Romero and Ventura, 2020). This project addresses all three: it works with one instructor's data on its own terms, treats multicollinearity as a first-class methodological problem, and uses explainable-ML techniques (permutation importance and SHAP values) to produce a ranked, interpretable account of which features contribute independently to prediction.

3. Methodology

3.1 Overall design

This project is a secondary-data quantitative study using anonymised CSV files supplied by the module instructor: an engagement file mapping each student ID to per-feature interaction counts, a current-cohort grades file, and a previous-cohort grades file. Because no new data is being collected from human participants, no participant recruitment strategy is required. Ethics approval is nonetheless required: even with anonymised secondary data, the project handles educational records and will therefore undergo a lowrisk ethics registration before analysis begins, alongside written confirmation from the module instructor of data-sharing permission, anonymisation, and secure local storage. Layering primary data collection on top would add ethics overhead, recruitment risk, and a longer timeline.

3.2 Why this approach, and not the alternatives

Three plausible alternative designs were considered and rejected.

Alternative 1: descriptive statistics and correlation only. The simplest defensible design would compute pairwise correlations between each feature and the grade outcome, then rank them. This was rejected because, by construction, correlation cannot separate independent feature effects from the shared 'total access' signal that the null hypothesis is built around. A correlation table would say everything matters, which is exactly what the brief warns against.

Alternative 2: a single regression model. Multiple linear regression with feature counts as predictors yields interpretable coefficients but was rejected as the sole approach for two reasons: the literature in Section 2.2 strongly suggests that the true relationship is non-linear and includes feature interactions that a linear model under-represents; and multicollinearity will make the coefficients unstable. Regression is therefore retained as a baseline (Activity 4) but not as the headline method.

Alternative 3: process or sequence mining. A different family of methods would model the order in which students use features over time. This suits LMS clickstream data, but the brief describes the data as aggregated engagement counts rather than time-ordered event logs, so sequence mining would require a richer dataset. It is noted as future work rather than the chosen method.

The chosen approach: three machine learning models (Random Forest, XGBoost, and a small Neural Network) fitted on the same engineered features and evaluated under the same protocol; captures nonlinearities and interactions that a regression cannot, produces feature-importance rankings that correlation cannot, and works with the data described in the brief.

3.3 Data preparation and feature engineering

All preprocessing will be performed in Python using pandas, scikit-learn (Pedregosa et al., 2011) and statsmodels. The pipeline begins with a data-quality audit: checking ID linkage between the engagement

and grade files, profiling missingness, and identifying outliers using the 1.5×IQR rule, with outliers treated on a case-by-case basis rather than dropped by default. A data dictionary will be maintained throughout.

Feature engineering converts raw counts into more informative variables: per-feature totals, engagement ratios (e.g., quiz attempts per session), and short-window rolling averages where the data support them. The shape of the engagement file determines what is possible: if it contains time-stamped events, rollingwindow features and time-aware cross-validation will be used; if it is aggregated at the studentfeature level with no time dimension, rolling features will be omitted and stratified k-fold cross-validation used. Total access count is retained as a separate variable because it is precisely the variable the null hypothesis privileges, and the regression in Activity 4 will include it as a control before any feature counts are added.

3.4 Modelling and evaluation

Three machine learning models will be fitted: a Random Forest classifier, an XGBoost classifier, and a small fully connected neural network. Random Forest and XGBoost are well established for tabular educational data and perform reliably on small-to-medium datasets with mixed feature types (Conijn et al., 2017; Saqr and Ló pez-Pernas, 2021); the neural network tests whether a more flexible architecture detects patterns that the tree-based models miss, with the caveat (R5) that small samples may render it unreliable.

The cross-validation strategy follows the data shape, as set out above. Hyperparameter tuning will use Bayesian Optimisation (Snoek, Larochelle and Adams, 2012), which is more efficient than grid search when each model fit is costly. The reported metrics will be macro-averaged F1 and ROC-AUC as primary measures (raw accuracy is misleading under class imbalance, R3), with accuracy, precision, and recall as secondary diagnostics.

Feature importance will be reported for the best-performing model using both permutation importance (Strobl et al., 2008), which is more robust than default tree-based importance measures in the presence of correlated predictors, and SHAP values (Lundberg and Lee, 2017). Convergence between methods strengthens confidence in the conclusions; divergence is diagnostically interesting and will be reported rather than suppressed.

3.5 Limitations of the chosen design

Three limitations are inherent. Observational data cannot support causal claims; the strongest the study can offer is feature effects that survive controls for the most plausible confounder (total access). The previous cohort baseline is not perfectly comparable; cohort-level differences cannot be attributed solely to the new features. Machine-learning feature importance is not the same as causal effect size; an important predictor may still be useless to manipulate in practice. Each shape how the findings should be interpreted. **4. Project Risk Analysis**

The risk register below identifies eight project-specific risks, each rated for likelihood and impact on a Low/Medium/High scale, with a concrete mitigation for each. The risks are specific to the analytical and practical realities of this project; they are not generic placeholders, although personal circumstances (R8) are also included, as contingency time has been built into the schedule.

Risk	Likelihood	Impact	Mitigation
R1(Data linkage failure):			

anonymised IDs in the engagement file may not map cleanly to the grade file, leaving rows unmatched.	Medium	High	Audit ID consistency in Week 2 before any modelling begins. Document the linkage rate; if more than 10% of records cannot be matched, raise this with the module convenor rather than silently dropping rows.
R2(Multicollinearity): Multicollinearity between engagement features (students who use one feature heavily tend to use all of them), making	High	High	Compute Variance Inflation Factors before regression. If VIF > 5 on multiple features, switch from standard regression to Lasso (Tibshirani, 1996), which performs feature selection under

Risk	Likelihood	Impact	Mitigation
individual feature effects hard to isolate.			collinearity. This is a pre-committed methodological decision, not a workaround.
R3(Class imbalance): Class imbalance in the grade outcome (e.g., few failing students), which inflates accuracy but hides poor minority-class recall.	High	Medium	Bin grades into reasonably balanced categories where possible. Report macro-averaged F1 and ROC-AUC as primary metrics rather than raw accuracy and apply class weighting in the Random Forest and XGBoost classifiers.
R4(Temporal leakage): Rolling averages or aggregated features accidentally use information from after the prediction point, inflating test performance.	Medium	High	Use TimeSeriesSplit cross-validation (Raschka and Mirjalili, 2019). Confine all aggregation windows to data available before each fold's split point, and document this explicitly in the analysis notebook.

R5(Sample size): Small effective sample size after cleaning, reducing statistical power and making the Neural Network unreliable.	Medium	Medium	Profile the dataset in Week 2 and report the final n. If the sample is small, prefer parsimonious models (regularised logistic regression, shallow Random Forest) over the Neural Network, which needs more data to fit reliably.
R6(Model performance): Models reach a performance ceiling and feature importances disagree across Random Forest, XGBoost, and the Neural Network.	Medium	Medium	Treat divergence as a finding, not a failure: report which features are stable across models, and which are not. Use permutation importance (Strobl et al., 2008), which is more robust under correlated predictors.
R7(Scope creep): the temptation to add new architectures, features, or analyses outside the agreed plan.	Medium	Low	Maintain a written scope document; any addition must displace something already planned. Weekly self-review against the Gantt chart.
Risk	**Likelihood**	**Impact**	**Mitigation**
R8(Personal circumstances): illness, work commitments, etc cause a multi-week delay.	Low	High	One-week buffer built into Weeks 15–16. Maintain weekly written progress notes so that, if Extenuating Circumstances need to be invoked, the evidence trail is already in place.

The two dominant risks are R2 (multicollinearity) and R3 (class imbalance), which are methodological. Their mitigations are pre-committed analytical decisions baked into the methodology; a switch to Lasso under high VIF, and a switch from raw accuracy to macro-F1 and ROC-AUC under imbalance, rather than reactive workarounds. The cumulative effect is to reduce residual risk to a level at which the project remains achievable even in the face of adverse events.

5. Gantt chart

The project runs for 17 weeks, from mid-May to mid-September 2026, fitting comfortably within the brief's recommended Spring start and mid-September finish. Ten activities have been identified, each with a

timeline and an explicit deliverable; four of these activities terminate at named milestones (M1–M4). The activity table below is the textual companion to the Gantt chart.

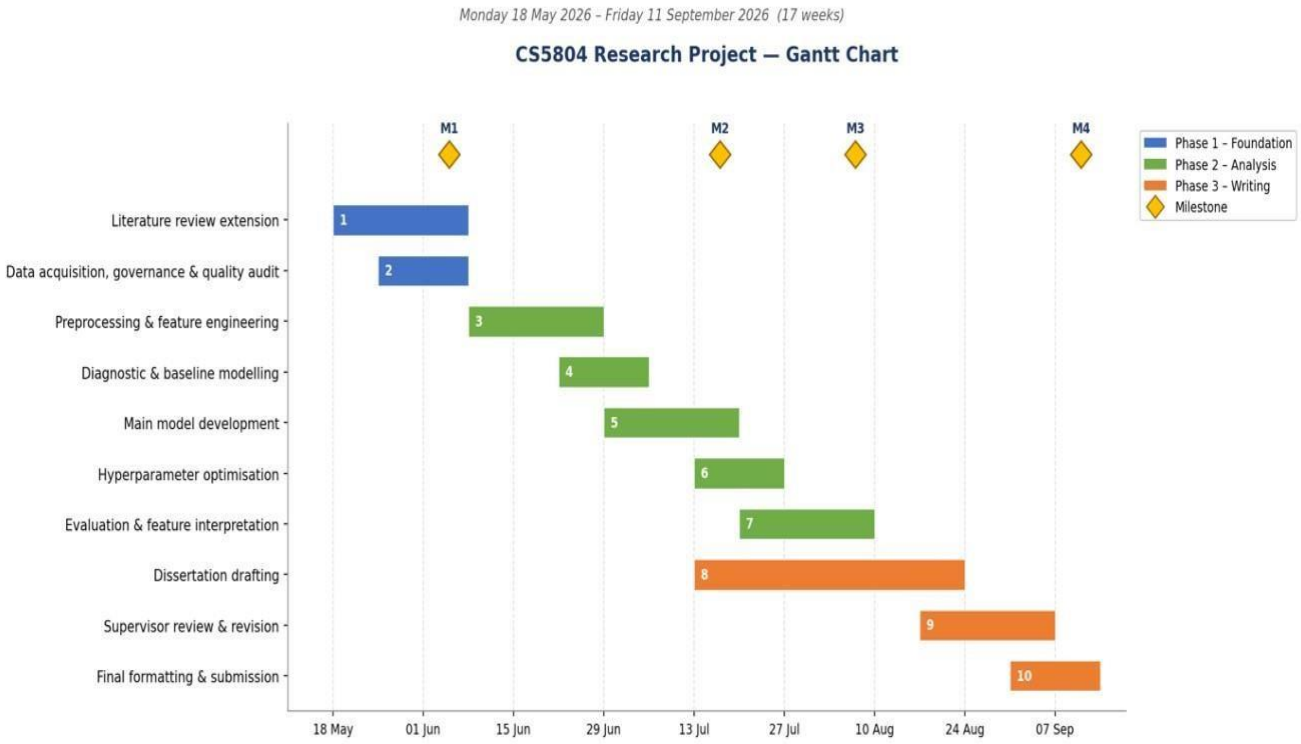


Figure 1: Gantt Char **Activity Descriptions:**

1. Literature review extension (Weeks 1–3). Extend the reading begun for this proposal: multicollinearity in LMS analytics, temporal engagement modelling, and explainable ML.
2. Data acquisition and quality audit (Week 2). Submit the low-risk ethics registration; obtain written confirmation of data-sharing permission, anonymisation, and secure storage. Build a data dictionary; audit ID linkage; profile missingness, outliers, and class balance; and confirm whether the data is time-stamped or aggregated. Milestone M1 (end of Week 3): dataset validated.
3. Preprocessing and feature engineering (Weeks 4–5). Handle missing values, encode categorical variables, scale counts, and engineer the features described in Section 3.3. Deliverable: a reproducible preprocessing notebook.
4. Diagnostic and baseline modelling (Weeks 6–7). Compute correlations and VIFs for the engineered features. Fit a regularised logistic regression as a transparent baseline for comparison with the more complex models.
5. Main model development (Weeks 7–9). Train Random Forest, XGBoost, and the small Neural Network using identical features, with cross-validation as specified in Section 3.3. Milestone M2 (end of Week 9): all three models are trained.
6. Hyperparameter optimisation (Weeks 9–11). Apply Bayesian optimisation to each model. Record the search space, prior, and best configuration to ensure reproducible tuning.
7. Evaluation and feature interpretation (Weeks 10–12). Compare models using macro-F1 and ROCAUC; compute permutation importance and SHAP values for the best model. Milestone M3 (end of Week 12): feature ranking complete.
8. Dissertation drafting (Weeks 9–13). Draft chapters in parallel with the analysis they describe. Deliverable: a complete first draft.
9. Supervisor review and revision (Weeks 13–15). Submit the full draft and incorporate feedback.

10. Final formatting and submission (Weeks 16–17). Cross-check citations; convert to PDF; submit. Milestone M4: dissertation submitted.

Three project-management principles underpin this plan. First, parallelism: the literature review and data audit run concurrently with early preprocessing, rather than sequentially. Second, controlled slack at the end: Weeks 13–17 combine supervisor revision, formatting, and contingency time. Third, milestone-based monitoring: the absence of M1, M2, M3, or M4 triggers a written replanning note rather than an unrecorded slide. These reflect the adaptive, feedback-driven planning that Saunders, Lewis and Thornhill (2019) argue is essential for research projects under fixed deadlines.

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